R-501A
RD REACTOR-I
CAPACITY: 43 M3
DIM: 3650 MM ID X 3750 MM T-T
DP(V//C): 1/FV//9 BAR(G)
DT(V//C): 170/5//200/5 °C
MOC(V/C): SS316L/SS316L
INS: HOT

A-501A
RD REACTOR-I AGITATOR
TYPE: TOP ENTRY
MOC: SS316L
MOTOR: XX KW

D-501A
TOLUENE WATER SEPARATION DRUM-I
CAPACITY: 3 M3
DIM: 1400 MM ID x 1800 MM T-T
DP: 1/FV BAR(G)
DT: 100/5° C
MOC: SS304
INS: NONE

P-1001A PID-1010 — FQC P1010A01 — FT P1001A01 — (NOTE-7)
FRESH ECH PID-1010 FROM P-1001A — SOL-101016-40-B2 NNF — 25 — 40xXX — C — XXx40 — XV P1001A01 (NOTE-12) FC (TYPE-1)

REC. ECH PID-0510 FROM P-508 — SOL-051008-80-B2 — 25 — XV P5080I (NOTE-12) FC (TYPE-1)

COLD DM WATER PID-1210 FROM HEADER — DM-121065-100-B1-C — XV R501A13 (NOTE-12) FC (TYPE-1)

HOT DM WATER PID-1210 FROM HEADER — DM-121051-100-B1-H — DM-050104-40-B1 — XV R501A07 (NOTE-12) FC (TYPE-1) — FT R501A04 — FQI R501A04

XV R501A14 NC FC (TYPE-1)

P-1002B PID-1012 — FQC P1002B01 — FT P1002B01 — XV P1002B01 FC (TYPE-1) (NOTE-12) — B1 B2

CAUSTIC SOLN. PID-1012 FROM P-1002B — NA-101210-40-B2-ET-H — 25 — 40xXX — C — XXx40 (NOTE-7) (NOTE-10)

CAUSTIC SOLN. PID-0509 TO D-507 — NA-050103-25-B2-ET-H

TOLUENE PID-0510 FROM P-506 — SOL-050104-80-B2 — 25 — XV P5060I (NOTE-12) FC (TYPE-1)

PHENOL PID-1005 FROM P-1008A/B — SOL-100504-80-B2-ET-H — XV P1006O1 (NOTE-12) FC (TYPE-1)

ECOROL PID-1006 FROM P-1010 — SOL-100610-80-B2 — 25 — XV P101001 (NOTE-12) FC (TYPE-1)

CRESOL PID-1005 FROM P-1011A/B — SOL-100516-80-B2-ET-H — XV P1011B01 (NOTE-12) FC (TYPE-1)

1,4 BD PID-1006 FROM P-1009 — SOL-100603-50-B2-ET-H — 25 — XV P100901 (NOTE-12) FC (TYPE-1)

PID-0502 TO PV-V501A01 — FI R501A01 — FT R501A01 — PIC R501A01 — PV R501A01 — 25xXX FC XXx50 — A1 B2 (HOLD-1)

N1 PID-1212 FROM HEADER — N1-121251-25-A1

SOL-050107-80-B2-ET-H — SOL-050101-40-B2-ET-H — NA-050102-40-B2-ET-H — NA-050101-40-B2-ET-H — PRO-050104-HOLD-B2

TRV R501A01 S.P.=9 BARG SIZE= 1D2 (NOTE-14) — XV R501A05 SAFE LOCATION — VENT TO — FC (TYPE-1) — 25 — PG R501A01 — PT R501A01 — PG R501A02

HS R501A02 — ZSCC R501A01 — S — TCCV R501A01 — 150xXX FO XX150 (HOLD-1) — TW R501A01 — 25 — CWR PID-1206 TO HEADER — CWR-050101-150-A1

TIC R501A01-C — TIC R501A01-H — H 150x80 — L — 150x80

A1 NONE A4 H — HS R501A01 — ZSCH R501A01

FI R501A02 — FT R501A02 — TCHV R501A01 — S — 80xXX (HOLD-1) — FC 80xXX

MPS PID-1208 FROM HEADER — MPS-120851-80-A4-H

CWS-120551-150-A1 — CWS PID-1205 FROM HEADER — NONE H — XV R501A03 FO (TYPE-1)

SM — T — MPC PID-1209 TO HEADER — MPC-050101-80-A1-H TWO PHASE FLOW — T

SM — 150x80 — FC (TYPE-1) — XV R501A04 — H PP — FC (TYPE-1) — XV R501A02 — A — WW-050104-50-A1-PP

N1 PID-1212 FROM HEADER — N1-121257-25-A1

WASTE WATER PID-0510 TO T-502 — WW-050101-150-B2-PP — 1:3.7

R-501A

C17A 50 — C17B 50 — C17C 50 — C17D 80 — R1 40 — R2 40 — R3A 80 — C23 50 — C9 50 — C18C 50 — C18B 50 — C18A 50 — C10 80 — C18D 80 — C14 150 — A4 A1 — (NOTE-4) — (NOTE-16) — (NOTE-15) — C16 80 — C6 100 — C3A 40 — C20 40/25 — C3B 40 — C3C A1 — C8 100 — C7 80 — C22 HOLD — C13 350 — C12 100 — C15 200 — C4 50 — C19 50 — C21 100 — R3B 80 — A2 600 — R4A/B 150 — R5 50 — C1 50 — R6 50 — C5 50

(NOTE-13) — VFD — HS — (TYPE-C) — M — 1,6HD

C-050103-50-B5 — NaH2PO4 PID-0509 FROM P-505

V-050101-350-B2-PP — VENT PID-0502 TO E-502A

SOL-050105-100-B2 — XV D501A03 FC (TYPE-1) — SOL-050202-100-B2 — GRAVITY FLOW — TOLUENE/WATER PID-0502 FROM E-502A/503A

XV D501A02 FC (TYPE-1) — V-050102-80-B2 — VENT PID-0502 TO E-503A

SOL-050101-100-B2 — SM — XV D501A01 FC (TYPE-1) — 300 MM — 25 — R2A/B 150 — C2 150/100 — C3 100 — V1 80 — C1 50 — C6 40 — R4A — R4B — R4C — R4D — TG D501A01 — R1 40 — C5 25 — C4 50 — R3B 80 — A1 600 — R3A 80 — LT D501A01 — INT — LIC D501A01 — H — L — GRAVITY FLOW

WW-050105-25-B2 — REACTION WATER PID-0114 TO D-170

FQI D501A01 — FT D501A01 — WW-050103-25-B2 GRAVITY FLOW — WW PID-0510 TO D-505

(NOTE-11) — D-501A — SOL-050102-50-B2 — LV D501A01 — 25 — 50xXX FC XXx50 — 25 — 50x25 — 25 — C (NOTE-7)

XV D501A04 — 25 — SOL-050106-100-B2 — FC (TYPE-1) — XV D501A05 — FC (TYPE-1) — SOL-050103-50-B2 — RECOVERED ECH PID-0510 TO D-502

GRAVITY FLOW — TOLUENE PID-0510 TO D-504

SM — (HOLD-2) S.P.=1 BARG — PSV R501A01 — PT R501A01 — PI R501A01 — (V) — PSE R501A01

PRO-050202-50-B2-H — WASTE POLYMER RECYCLE PID-0502 FROM P-510A

100x50 — PRO-051001-50-B2-H — RESIN PID-0510 FROM P-507

PRO-050311-100-B2-H — RESIN PID-0503 FROM P-501A

XV R501A11 (NOTE-15) (NOTE-12) — CT-050101-25-B5 — MIN. — 50x25 — 25 — LT R501A01 — LI R501A01 H L — SNCL4 — BF3

(NOTE-2) — SM — (NOTE-6) CAMERA — XV R501A10 (NOTE-12) FC (TYPE-1) — PP H

XV R501A06 FC (TYPE-1) — PRO-050101-50-B2-H — WASTE POLYMER PID-0502 TO D-503A

XV R501A12 FC (TYPE-1) (NOTE-3) — PRO-050103-150-B2-H — RESIN+WATER PID-0503 TO P-501A

GENERAL NOTES:

1. ALL VAPOR LINES, SLURRY LINES AND GRAVITY LINES SHOULD HAVE MINIMUM BENDS AND NO LOOPS.
2. PROVIDE PROPER EARTHING AND BONDING FOR THE HYDROCARBON LINES.
3. DELETED.
4. DRAIN TO BE PROVIDED AT THE LOWEST POINT AND VENT TO BE PROVIDED AT THE HIGHEST POINT.
5. REFER LEGEND SHEET FOR ABBREVIATIONS, SYMBOL, ON-OFF VALVE DETAILS, MOTOR/VFD AND STEAM TRAP DETAILS.
   DWG NO. EPAM01-PR-PID-1000, EPAM01-PR-PID-1001, EPAM01-PR-PID-1002.
6. FOR LINES WITH SLOPE, SLOPE TO BE PROVIDED IN SUCH A WAY THAT THE LINE SHOULD BE DRAINED ON EITHER SIDE.
7. PROPOSED SPECIAL PIPING ITEM. TO BE FINALISED DURING DETAIL ENGINEERING.
8. TYPICAL XV SYMBOL ON PID TO BE READ AS BELOW.

XV XXXX (TYPE-1,2,3) — HS XXXX — XV XXXX (TYPE-1,2,3)

9. FOR ALL INLINE INSTRUMENTS, UPSTREAM AND DOWNSTREAM DISTANCE TO BE CONSIDERED IN PIPING.
10. ORIFICE TO BE INSTALLED IN HORIZONTAL LINE ONLY.
11. INSTRUMENT ISOLATION VALVES & DRAIN/VENT REQUIREMENT TO BE UPDATED DURING DETAILED DESIGN ALONG WITH THE P&ID.

NOTES:

1. AGITATOR MOTOR WILL TRIP FROM DCS IN CASE OF MECHANICAL SEAL LEAKAGE.
2. SAMPLE POINT PROVISION FOR R-501A, SAMPLE DRAIN TO BE COLLECTED MANUALLY.
3. TO BE AT MINIMUM DISTANCE WITH TAPPING FROM R-502A.
4. DIP PIPE SHOULD BE REMOVABLE.
5. DELETED.
6. CCTV VISIBILITY TO BE PROVIDED IN DCS ROOM.
7. FT-P1001B01, FT-P1002B01, FT-D501A01 SHALL BE MOUNTED ON A VERTICAL LINE.
8. FLUSH BOTTOM VALVE TO BE INSTALLED WITH A HANDWHEEL.
9. UTILITY PIPING SHALL BE SELF DRAINING AND NO POCKETS TO ASSIST QUICK DRAINING.
10. USE LUBRICATED PLUG VALVES & PROVIDE FLANGE GUARDS FOR CAUSTIC LINES.
11. ENSURE MIN. 11M BAROMETRIC LEG FROM D-501A TO D-502/ D-504/ D-505 RESPECTIVELY.
12. XV TO BE AT A MINIMUM DISTANCE FROM R-501A.
13. PROVISION FOR DIP PIPE CLEANING.
14. FUNNEL ARRANGEMENT & ROUTING TO BE DONE DURING DETAILED ENGINEERING.
15. XV-R501A11 TO BE PTFE LINED.
16. CAUSTIC DIP PIPE SHOULD BE LOCATED AWAY FROM COIL.

HOLDS:

1. CONTROL VALVE SIZE.
2. PSV SIZE.

No.	DATE	REVISION	DRAWN	CHD	REVD	APPD
0	01-JUN-26	ISSUED FOR DETAIL ENGG.	DVS	NC	DP	KV
B	29-MAY-26	ISSUED FOR DESIGN	DVS	NC	DP	KV
A	27-MAR-26	ISSUED FOR APPROVAL	DVS	NC	DP	KV

Rasayan Engineering Solutions

CLIENT: EPIGRAL Limited

PROJECT: BASIC ENGINEERING FOR EPAM PROJECT

TITLE: PIPING & INSTRUMENTATION DIAGRAM RD REACTOR SECTION (RD) TRAIN-I

SCALE : NTS | DWG.NO.: EPAM01-PR-PID-0501 | SHT: 1 OF 1 | REV. 0